The Decreasing Universe Theory as an Alternative Program for Cosmological Interpretation: Redshift, Time, Apparent Dark Matter, and Observational Tests

Abstract

This article critically reviews a set of papers devoted to the so-called **Decreasing Universe Theory** — DUT —, a proposal according to which gravitational fields produce a continuous contraction of space, of objects, and of the measuring instruments located within it. The review aims to analyze the internal coherence of this theoretical program and to examine how its papers mobilize a common hypothesis to reinterpret phenomena usually associated with cosmic expansion, dark energy, dark matter, cosmological redshift, the time dilation observed in Type Ia supernovae, the wavelength of the cosmic microwave background, and observational correlations involving mass, galactic age, and redshift. The corpus analyzed includes theoretical papers, cosmological applications, and preliminary empirical tests. The analysis shows that DUT exhibits a strong unifying ambition: it seeks to derive, from a single premise, effects attributed in the standard cosmological model to different entities or mechanisms. At the same time, the documents themselves reveal important limitations, especially the need for greater mathematical formalization of the relationship between contraction, mass, gravitational field, and time, as well as the need for independent tests on large samples, with rigorous control of peculiar velocities, projection effects, total mass, and stellar age. It is concluded that DUT, as presented in the documents analyzed, should be understood as a speculative yet testable theoretical program, whose main contribution is to formulate alternative observational hypotheses, but whose scientific acceptance would depend on independent empirical validation and systematic quantitative comparison with competing models.

Keywords: Decreasing Universe Theory; redshift; dark matter; dark energy; Tully-Fisher relation; Type Ia supernovae; alternative cosmology.

1. Introduction

Contemporary cosmology is structured, to a large extent, around the ΛCDM model, which combines cosmic expansion, the cosmological constant or dark energy, cold dark matter, and general relativity to explain a wide range of observations. The documents analyzed here propose an alternative interpretation, called the Decreasing Universe Theory — DUT —, whose central principle is that the gravitational field continuously contracts space and everything within it. According to this hypothesis, the contraction would be extremely slow on local scales but cumulative on cosmological scales, potentially becoming perceptible in long-duration astronomical phenomena. In the article on time dilation in Type Ia supernovae, for example, it is stated that contraction within Earth's gravitational field would be on the order of 7% per billion years, and that observers located in regions of negligible gravity could be treated as “comoving” observers, while observers immersed in gravitational fields would be “proper” observers.

DUT's proposal shifts the interpretive focus: what appears as accelerated expansion of the universe could result from the contraction of local measurement standards; what appears as cosmological redshift could be partly an effect of comparing physical systems subjected to different contraction histories; and what appears as anomalous orbital velocity in galaxies could be reinterpreted as a consequence of differential gravitational contraction. The article on Type Ia supernovae itself lists three factors that would contribute to galactic redshift under DUT: mass, age, and distance. More massive galaxies would tend to exhibit lower redshift under comparable conditions; older galaxies would also tend to exhibit lower redshift; and more distant objects would tend to exhibit higher redshift due to the longer photon propagation time to the observer.

The purpose of this article is to critically analyze, in Mini Review format, how the set of attached documents constructs this hypothesis and applies it to different cosmological and galactic problems. The analysis does not aim to demonstrate the external validity of DUT or to replace a systematic review of the international cosmological literature. Its scope is more limited: to reconstruct the program's internal logic, identify its main applications, examine the proposed observational tests, and point out methodological and theoretical gaps that would need to be overcome for the theory to be evaluated according to more demanding academic standards.

The review is organized into three main sections. The first examines the foundations of gravitational contraction and its applications to redshift, distance, cosmological time, Type Ia supernovae, and the cosmic microwave background. The second discusses applications to galactic dynamics, especially rotation curves and the Tully-Fisher relation. The third analyzes the observational tests involving mass, galactic age, and redshift, highlighting favorable results, mixed results, and methodological limitations.

2. Review Methodology

This work adopts a critical narrative review methodology, appropriate when the goal is not to perform a statistical meta-analysis but to conceptually organize a delimited set of texts. The documents attached by the user were analyzed, all centered on the Decreasing Universe Theory or its applications. The corpus includes papers on distance-redshift, time dilation in Type Ia supernovae, the cosmic microwave background and the age of the universe, the apparent dark matter effect, the Tully-Fisher relation, mass-redshift correlation, galactic age and redshift, as well as tests on subgroups of galaxies in the Coma Cluster.

The reading was guided by four questions: first, what is DUT's central hypothesis and how is it mathematically formulated? Second, how is this hypothesis applied to cosmological phenomena usually explained by expansion, dark energy, or relativistic effects? Third, how does the theory seek to reinterpret phenomena associated with dark matter and galactic dynamics? Fourth, what observational tests are proposed, and what limitations do the documents themselves acknowledge?

Because this is a review based exclusively on the documents provided, statements about DUT are presented as internal formulations of the corpus analyzed. When the texts claim to challenge or refute the ΛCDM model, this claim is treated as the documents' position, not as a conclusion established by the scientific community. This distinction is methodologically important because the texts themselves acknowledge, at certain points, the existence of alternative explanations compatible with ΛCDM, especially in the case of redshift differences by galactic morphology and the effects of peculiar velocities in clusters.

3. Foundations of Gravitational Contraction: Distance, Redshift, and Cosmological Time

DUT's core hypothesis is that the gravitational field continuously contracts space and the objects contained within it. In epistemological terms, this shifts the cosmological problem to the relationship between observer, measuring instrument, and local gravitational history. An observer immersed in a gravitational field would not directly perceive the contraction of their own measurement standards, because their instruments, their atoms, and local objects would shrink together. The effect would become perceptible only in comparison with regions or phenomena whose gravitational history was distinct. In the article on Type Ia supernovae, this distinction is expressed through the concepts of the comoving observer, located in a region of negligible gravity, and the proper observer, situated in a gravitational environment.

This distinction between reference frames allows redshift to be reinterpreted. Rather than being understood exclusively as a product of the expansion of space or of recession velocities, it would, within DUT's framework, also be an effect of the contraction of local measurement standards. The more distant the observed galaxy, the longer the photon's travel time; during that interval, the observer and their instruments would continue to contract, such that the received wavelength would be measured as relatively longer. The appearance of accelerated expansion would arise, in this interpretation, from the continuous contraction of local distance standards.

The article on distance as a function of redshift develops this intuition into a mathematical formulation, seeking to derive a relationship between apparent distance, comoving distance, and redshift. In later documents, this structure is taken up again in the form of the so-called Jocaxian factor, associated with an exponential function of time and, in certain contexts, with the Hubble parameter. The article on the Tully-Fisher relation, in reconstructing DUT's foundations, presents the formula for distance as measured by a contracting reference frame and links it to Hubble's law by differentiation, producing a relationship between apparent velocity and distance.

DUT also seeks to extend its hypothesis to the problem of time. The article on time dilation in Type Ia supernovae argues that, if units of distance contract, units of time must also be affected, since the physical definition of the second depends on atomic processes. From the comparison between units of time in comoving and proper reference frames, the text derives a relation in which the numerical value of time measured in the proper reference frame grows by a factor associated with contraction. According to the article, this formulation would explain the cosmological time dilation observed in Type Ia supernovae, whose apparent duration increases by the factor related to redshift.

The application to the CMB represents an even more ambitious extension. The article on the cosmic microwave background and the age of the universe uses DUT to estimate the cosmic age by comparing the observed average wavelength of the CMB with a wavelength that would be produced today in a laboratory by processes similar to those of a plasma at around 3000 K. The text calculates a laboratory wavelength on the order of 5×10^{-6} m, compares it with the observed average CMB wavelength of approximately 1.44×10^{-3} m, and obtains an estimate of 82 billion years for the age of the universe within DUT's framework.

This estimate, however, should be interpreted with caution. The article itself acknowledges that the calculation assumes a contraction history associated with a gravitational field similar to the present one, even though the galaxy did not exist in its current configuration from the outset. For this reason, the text states that the calculated age should be understood as a lower bound, not as a definitive value. It is also noted that the article does not necessarily reject the possibility of a Big Bang, but redefines its status, treating the origin of the universe as an open question within the model.

The main strength of this first dimension of DUT lies in its attempt to unify distance, redshift, and time under a single physical hypothesis. Its main weakness, however, lies in the need to more rigorously formalize the relationship between contraction, gravitational field intensity, mass, exposure time, and cosmological evolution. Without this formalization, the theory remains conceptually suggestive but still insufficiently determined for broad quantitative comparison with the standard model.

4. Galactic Dynamics: Rotation Curves, Apparent Dark Matter, and the Tully-Fisher Relation

The second front of DUT's development concerns galactic dynamics. The article *Decreasing Universe: The 'Dark Matter' Effect* states that the gravitational contraction hypothesis could explain two phenomena usually treated separately: apparent accelerated expansion, without invoking dark energy, and anomalous orbital velocities of stars in galaxies, without invoking dark matter. The central argument is that differential gravitational contraction, varying with the intensity of the gravitational field in different regions of the galaxy, would alter the wavelength of photons emitted by stars located at different distances from the galactic center. Since stellar velocities are inferred from redshift, this difference could be mistakenly interpreted as a higher orbital velocity.

In the case of rotation curves, the article takes the galaxy Messier 33 as an example. The text states that the anomalous velocity of stars and gas in the outer regions of galaxies is traditionally explained by dark matter, since the apparent velocity does not decrease as expected under a simple Newtonian reading. DUT proposes that this behavior could be reconstructed by considering that photons emitted in different regions of the galaxy would undergo distinct contraction effects before being observed. The article develops an equation for apparent velocity as a function of the galaxy's mass, the star's radial distance from the center, the gravitational constant, the Hubble constant, and the Jocaxian factor.

The formulation culminates in an expression for the apparent galactic rotation curve, according to which the velocity inferred on Earth would result not only from the actual orbital velocity but also from the difference between the contraction of the emitting galaxy and the contraction of the Milky Way. The text states that this equation reproduces the observed rotation curve of Messier 33, traditionally attributed to dark matter.

From the standpoint of prudent academic writing, it is important to avoid the strong claim that dark matter has been eliminated. The more rigorous formulation is to say that, in the documents analyzed, DUT proposes an **alternative interpretation** for observables normally associated with dark matter. This distinction is essential because dark matter, in the standard cosmological model, is inferred not only from galactic rotation curves but from multiple classes of observations. Since the documents provided focus mainly on rotation curves, Tully-Fisher, and mass-redshift correlations, the conclusion should remain restricted to that scope.

The article on the Tully-Fisher relation represents an attempt to generalize DUT's application beyond a specific galactic case. The Tully-Fisher relation is presented in the document as an empirical law according to which the mass or luminosity of a spiral galaxy is proportional to the fourth power of its rotational velocity, $M \propto V^4$. The article claims to derive this relation from DUT, using the galactic rotation equation proposed in the article on the dark matter effect, together with two astrophysical approximations: an approximately constant mass-to-light ratio and Freeman's Law, understood as approximate uniformity of central surface brightness in spiral galaxies.

The derivation of the Tully-Fisher relation, as presented in the document, depends on expressing the galaxy's mass as a function of its area and an approximately constant average surface density. From this approximation, the article rewrites the rotation equation in terms of surface density and arrives at the proportionality $M \propto V^4$. In its conclusion, the text states that the Tully-Fisher relation can be derived without recourse to dark matter or an exotic component, and that this would reinforce DUT's ability to reproduce empirical regularities normally attributed to the standard model.

This section of the DUT program is theoretically relevant because it attempts to show that the contraction hypothesis not only explains an isolated phenomenon but can reproduce a general empirical relation. However, precisely because it depends on auxiliary hypotheses about spiral galaxies, it requires additional tests. It would be necessary to evaluate the derivation on low-surface-brightness galaxies, dwarf galaxies, systems with non-homogeneous mass distributions, galaxies with different evolutionary histories, and independent samples. Without these tests, the derivation remains interesting as a theoretical exercise but not yet sufficient to replace competing explanations.

5. Observational Tests: Mass, Galactic Age, and Redshift

The third dimension of the documents analyzed is the attempt to turn DUT into an empirically testable theory. This step is crucial, since an alternative cosmological theory can only be scientifically evaluated if it produces observational predictions distinguishable from those of competing models. The most recurrent prediction in the documents is the following: under comparable distance conditions, more massive galaxies should exhibit lower redshift; and, under comparable mass and distance conditions, older galaxies should also exhibit lower redshift.

The article *Decreasing Universe: Redshifts and Distance Data Refute the L-CDM Model* explicitly formulates this prediction in comparative terms. According to the text, while the ΛCDM model would interpret greater mass as associated with greater gravitational redshift, DUT predicts the opposite: more massive galaxies, at the same distance, should exhibit lower average redshift, since more intense gravitational contraction would reduce the wavelength of emitted photons. The article presents this hypothesis as a potentially falsifiable test.

The same work presents a table of 50 galaxy pairs, with approximate distance, estimated mass, and redshift, and states that 47 of the 50 pairs satisfy the condition predicted by DUT: at the same distance, the more massive galaxy would exhibit lower redshift. The text interprets this result as favorable to DUT and contrary to ΛCDM.

From a methodological standpoint, however, this result calls for caution. The article states that the data were obtained by querying an artificial intelligence tool, and the excerpt analyzed does not provide a detailed description of an astronomical catalog, independent selection criteria, mass uncertainties, distance uncertainties, or statistical error treatment. For a specialized academic community, these points would be decisive. The prediction is clear and therefore interesting; but its validation would require reconstructing the data from verifiable astronomical catalogs, with distances obtained by methods independent of redshift, masses estimated by homogeneous criteria, and adequate statistical treatment.

The galactic age hypothesis is developed in the article *The Age of Galaxies as a Determining Factor for Redshift*. The text starts from the idea that gravitational contraction is cumulative over time; therefore, an older galaxy would have accumulated greater contraction and should exhibit lower redshift than a younger galaxy under similar conditions. To test this hypothesis, the article uses morphology as a proxy for age: elliptical and lenticular galaxies are treated as old populations, while spiral galaxies are treated as younger populations with active star formation.

The article presents average redshift differences in clusters such as Virgo, Coma, and Centaurus. In all the cases listed, elliptical/lenticular galaxies appear with lower average redshift than spiral galaxies. The text interprets this pattern as aligned with DUT's qualitative prediction. However, the document itself acknowledges that the ΛCDM model offers a widely accepted alternative explanation: the difference could result from morphological segregation, peculiar velocities, and cluster dynamics. Elliptical galaxies, being older and more relaxed, would tend to occupy more central regions, while spirals could be in wider orbits or in the process of infall into the cluster, producing a higher average Doppler redshift.

This acknowledgment is methodologically important. It shows that the same observation can be compatible with two different interpretations. For DUT's interpretation to prevail, it would be necessary to demonstrate that the temporal contraction effect dominates over orbital dynamics. The article itself states, further on, that DUT's interpretation would require a more robust mathematical formalization of the relationship between lifetime, contraction, and redshift, capable of overcoming the established dynamical explanation.

The most methodologically refined study in the corpus is the article on mass-redshift correlation in subgroups with controlled radial velocities. In it, the sample is restricted to 20 elliptical/lenticular galaxies from the Coma Cluster, seeking to control for distance and stellar age. The set is then divided into subgroups with a redshift variation of approximately 150 km/s, aiming to reduce the noise produced by peculiar velocities. DUT's prediction is a negative correlation between stellar mass and observed redshift.

The results are mixed. Two subgroups show a negative correlation — one with a strong correlation of −0.8477 and another with a moderate correlation of −0.6789 — while another shows a perfect positive correlation, but with only two galaxies, and another shows a weak positive correlation. The article itself interprets the results as partially favorable to DUT but acknowledges that the absence of a universal negative correlation indicates that the proposed effect would not be the only factor at play.

The value of this study lies less in offering a definitive demonstration and more in proposing a variable-isolation strategy. By controlling for morphology, distance, and radial velocity, the article attempts to reduce factors that could mask the predicted correlation. At the same time, its limitations are made explicit: small subgroup size, the use of stellar mass as an imperfect approximation of total mass, possible projection effects, and differences in depth within the cluster. The text concludes that the negative correlation appears in controlled subgroups, but that the mass-dependent contraction mechanism still requires further investigation and mathematical formalization.

This observational section is the most important for the future academic evaluation of DUT. It shows that the theory can generate testable predictions, but it also makes clear that its current results are still preliminary. The theory would advance considerably if its tests were replicated using public catalogs, transparent criteria, larger samples, uncertainty analysis, Bayesian or frequentist comparison with alternative models, and ex ante quantitative prediction.

6. Discussion

Taken together, the documents present DUT as a program of conceptual unification. Its ambition is to replace multiple hypotheses or entities — dark energy, dark matter, accelerated expansion, and certain effects attributed to local dynamics — with a single physical hypothesis: the continuous gravitational contraction of space and of measurement standards. This ambition appears clearly in both the cosmological articles and the articles on galactic dynamics. The article on the dark matter effect, for example, states that the model would provide a unified framework for apparent cosmic acceleration and for anomalous rotation curves.

DUT's main virtue, in the corpus analyzed, is its attempt to formulate falsifiable predictions. The “greater mass, lower redshift” relation, the “greater age, lower redshift” hypothesis, and the tests on subgroups with controlled radial velocity are examples of propositions that can, in principle, be submitted to empirical evaluation. This is a positive aspect of the program, as it shifts it from a purely speculative formulation to a field in which its results can be confronted with observational data.

The main weakness lies in the transition from a qualitative hypothesis to a fully specified quantitative model. DUT states that contraction depends on the intensity of the gravitational field, mass, and exposure time, but the documents still leave gaps in the general formalization of this dependency. The introduction of auxiliary hypotheses, such as the inverse proportionality between contraction time and gravitational field intensity, allows the derivations to proceed, but needs to be justified, tested, and integrated into a more general mathematical framework.

Another problem is the need to distinguish contraction effects from effects already known in observational astrophysics. Redshift differences in clusters may arise from peculiar velocities, morphological segregation, depth within the cluster, and infall dynamics; rotation curves involve baryonic mass distribution, gas, halo, and galaxy inclination; relations such as Tully-Fisher depend on sample selection, photometric band, stellar mass, baryonic mass, and intrinsic scatter. The documents acknowledge part of these difficulties, especially in the case of morphological differences and the Coma subgroups.

Therefore, the most defensible academic contribution of DUT, at this stage, is not the definitive refutation of ΛCDM, but the formulation of an alternative program that produces observational hypotheses and calls for specific tests. Appropriate scientific language should avoid absolute conclusions and favor formulations such as “the preliminary data are compatible with,” “the results suggest,” “the hypothesis requires independent verification,” or “the model proposes an alternative interpretation.” This caution brings the article closer to the evaluation standards of international journals and reduces the risk of conclusions stronger than the data allow.

7. Conclusion

The analysis of the documents shows that the Decreasing Universe Theory is built around a simple and far-reaching hypothesis: gravitational fields would produce continuous contraction of space, matter, and measuring instruments. From this premise, the texts seek to reinterpret phenomena across different scales: cosmological redshift, apparent distance, time dilation in Type Ia supernovae, the wavelength of the CMB, galactic rotation curves, the Tully-Fisher relation, and correlations between mass, age, and redshift.

The program shows internal coherence insofar as it applies the same hypothesis to various problems. Its strength lies in its unifying ambition and in the formulation of potentially falsifiable predictions. The articles on mass-redshift, age-redshift, and Coma subgroups indicate an effort to make the theory empirically testable, which is an indispensable aspect of any alternative cosmological proposal.

However, the documents also reveal significant limitations. DUT still needs a more complete mathematical formulation of the relationship between contraction, mass, gravitational field, time, and photon emission/propagation. It also needs to be tested against independent astronomical catalogs, larger samples, rigorous control of dynamical variables, and quantitative comparison with competing models. The evidence presented is suggestive within the theory's own interpretive framework, but is not yet sufficient to establish a refutation of ΛCDM or a replacement of the prevailing cosmological paradigm.

Thus, the most balanced conclusion is that DUT should be understood as an alternative theoretical program under development. Its academic relevance will depend on its ability to convert its hypotheses into precise, replicable quantitative predictions capable of unambiguously distinguishing gravitational contraction effects from already-known dynamical effects. The most promising path forward involves building more robust mathematical models, reanalyzing large observational databases, and publishing independent tests that allow the assessment, under more rigorous statistical and physical criteria, of whether the patterns predicted by DUT persist when subjected to the scrutiny of the scientific community.

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